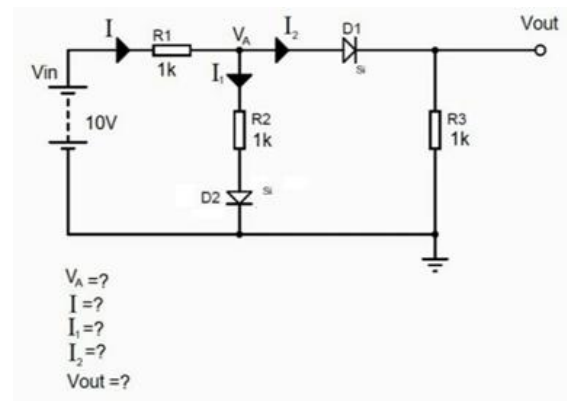
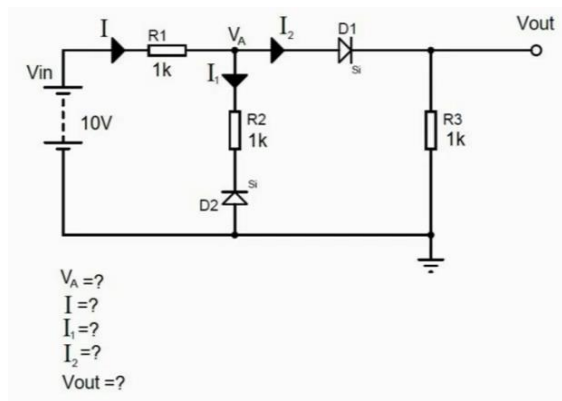
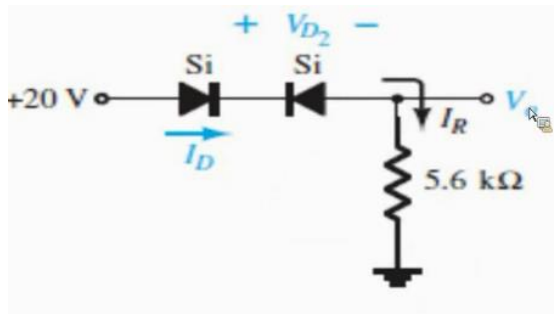
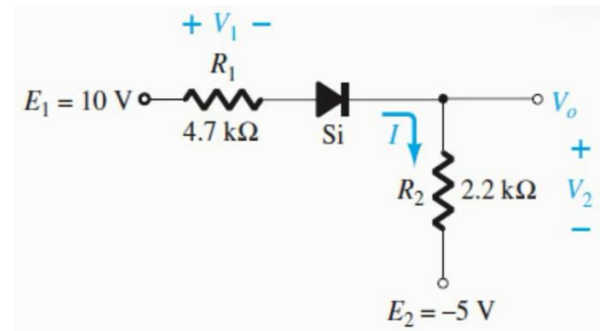


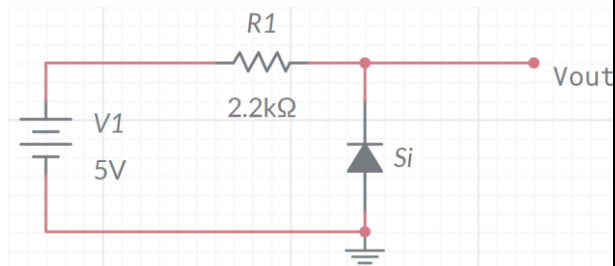




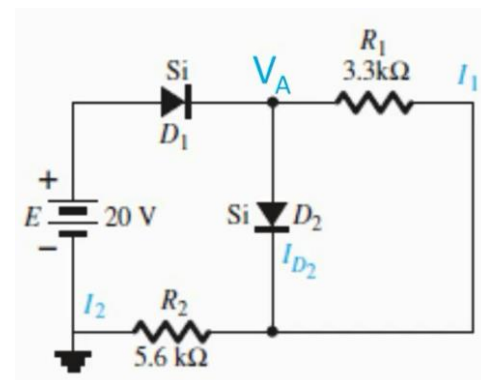
$$V_{out} = ?$$



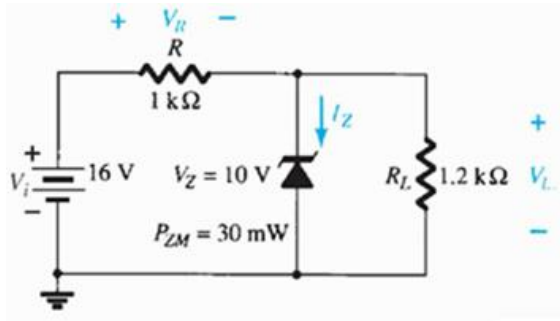
$$V_{out} = ?$$



$$V_{out} = ?$$



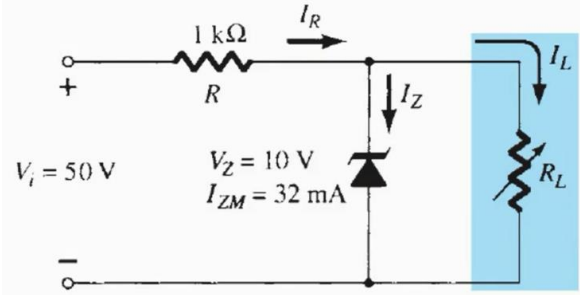
$$I_1, I_{D2}, I_2 = ?$$



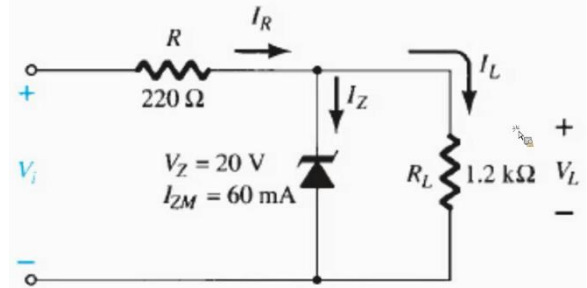
V_L , V_R , I_Z , and P_Z

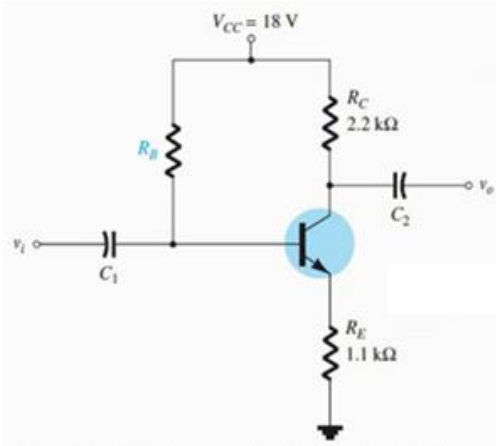
Direnç üzerine düşen gerilimi, yük üzerine düşen gerilimi, zener akımını ve gücü hesaplayınız. Ardından yük 3k iken işlemleri tekrarlayınız.

Yükten geçen akımın sınır değerlerini ve yükün sınır değerlerini bulunuz.

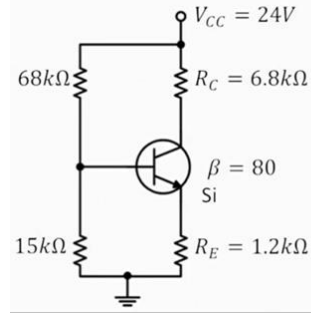
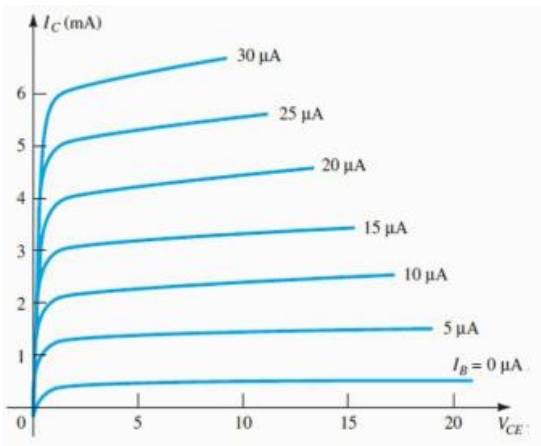


Giriş geriliminin alabileceği değerlerin sınırlarını bulunuz.



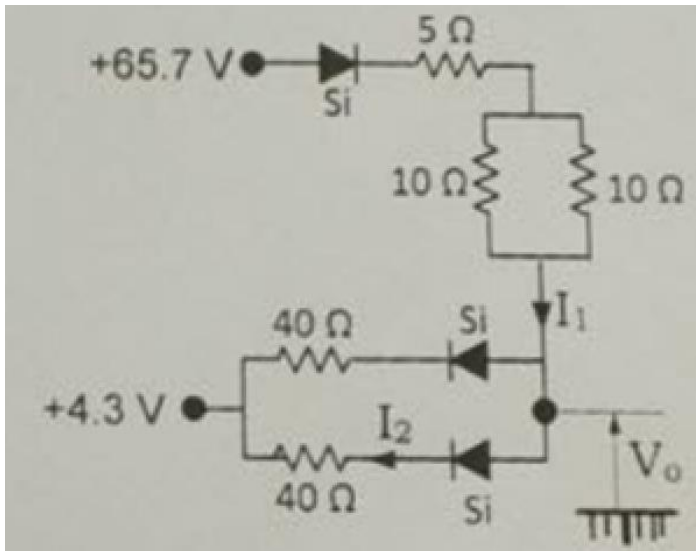


$\beta = 220$, $I_{CQ} = 3.3 \text{ mA}$ ise V_B , V_C , V_E , V_{CE} , I_B , I_C ve I_E değerlerini bularak, yük doğrusunu alttaki grafik üzerinde gösteriniz.

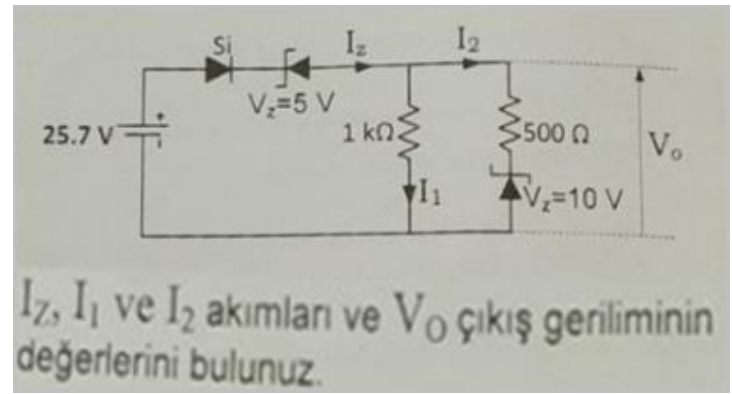


Şekildeki devrede V_B , V_C , V_E , V_{CE} , I_B , I_C ve I_E değerlerini

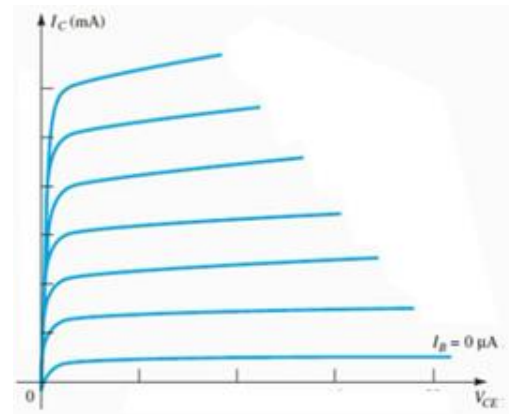
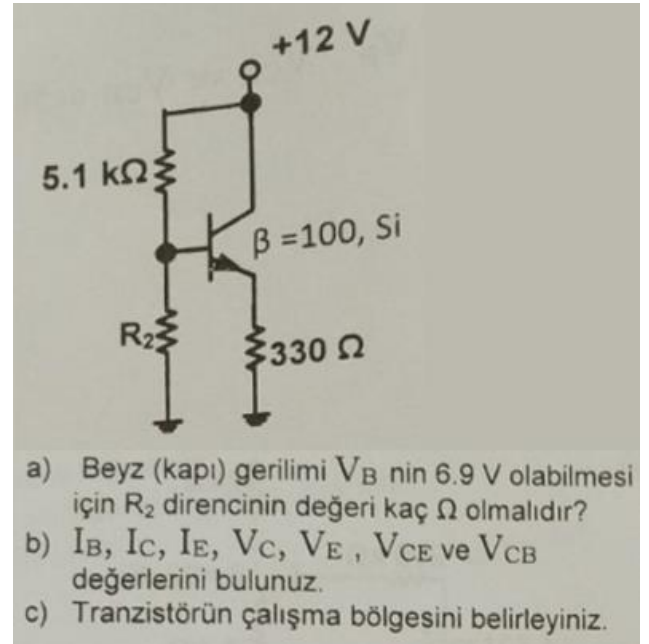
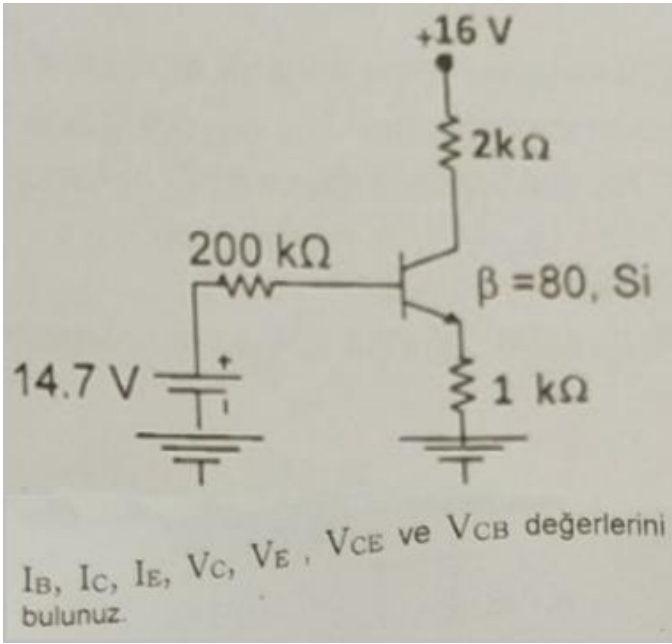
- Yüklemeye etkisini yok sayarak
♦ Yaklaşık Analiz ile ♦
- Yüklemeye etkisini hesaba katarak (tam analiz) bulunuz ve sonuçları kıyaslayınız.
♦ Tam Analiz ile ♦

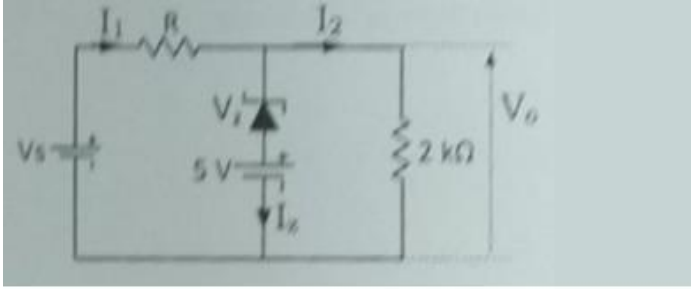


I_1 ve I_2 akımlarının ve V_0 çıkış geriliminin değerlerini bulunuz.

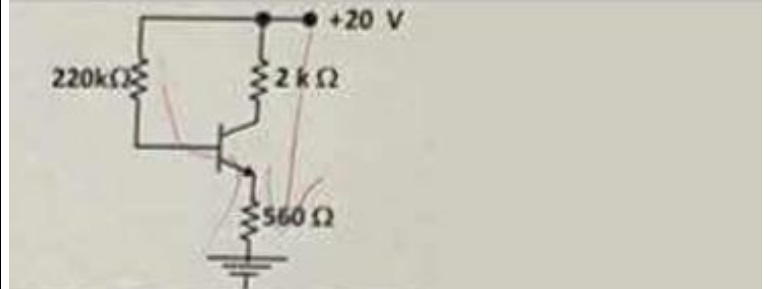


I_z , I_1 ve I_2 akımları ve V_0 çıkış geriliminin değerlerini bulunuz.

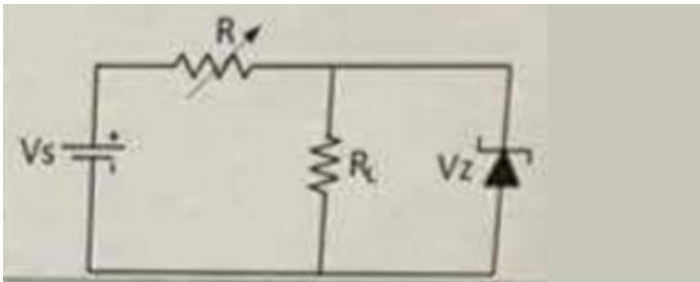




$V_S = 26V, V_Z = 15V, I_Z = 10mA$ olarak verilmiştir.
 $R = ?, I_1 = ?, I_2 = ?, V_O = ?$



- I_B, I_C ve I_E akımları ve V_B, V_C, V_E ve V_{CE} gerilimlerini bulunuz. $\beta = 80$.
- Tranzistörün çalışma bölgesini belirleyiniz.



$V_S = 30V, V_Z = 20V, R_L = 200\Omega, I_{Z_{min}} = 25mA, I_{Z_{max}} = 150mA$
 $R_{max} = ?, R_{min} = ?$

